

Qn		Answers	Marks
1	a	units of $v^2 = (\text{m s}^{-1})^2 = \text{m s}^{-2}$ units of $u^2 = (\text{m s}^{-1})^2 = \text{m s}^{-2}$ units of $2as = \text{m s}^{-2} \times \text{m} = \text{m s}^{-2}$ Since the units of all the terms are the same, the equation is homogenous. units of v^2, u^2, and $2as$ presentation and conclusion	B1 B1
	b	mass = (length $\times$ width $\times$ height) $\times$ density $= (1.99 \times 1.95 \times 2.02) \times 1.24$ $= 9.7199$ $= 9.71 \text{ g}$ By calculating percentage / fractional uncertainty: $\frac{\Delta m}{m} = \frac{\Delta \ell}{\ell} + \frac{\Delta w}{w} + \frac{\Delta h}{h} + \frac{\Delta \rho}{\rho}$ $= 0.01 + 0.01 + 0.01 + \frac{0.05}{1.24}$ $= 0.070323$ $\Delta m = 0.070323 \times 9.7199 [* \text{ M1}]$ $= 0.68353 = 0.7 \text{ g}$ mass = $(9.7 \pm 0.7) \text{ g}$ Alternatively, max-mean method: $m_{\text{max}} = (1.99 \times 1.01)(1.95 \times 1.01)(2.02 \times 1.01)(1.29) = 10.418 \text{ g}$ $\Delta m = m_{\text{max}} - m_{\text{mean}} = 10.418 - 9.7199 = 0.7 \text{ g}$	C1 C1 C1 A1 (C1) (C1)

2	2	Use of $mgh = \frac{1}{2} m v^2$ and makes h subject	
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